

Johnson C. Smith University
Introductory Linear Algebra
MTH 336(A) | 9AM-9:50AM SHB 303 | Fall 2013

Instructor Information

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Office hours:

10:00 AM–11:00 AM	MWF	
12:00 PM–1:00 PM	MW	
12:00 PM–1:00 PM	F	(by appointment only)
1:00 PM–2:00 PM	MW	(by appointment only)
11:00 AM–12:00 PM	TR	(by appointment only)

Course Description

This is a **3 credit hour** course; Systems of linear equations, vector spaces, and linear transformations, plus theory and applications of matrices and determinants. Prerequisite: MTH 231 or consent of Department. Prerequisite: MTH 231 (Calculus I).

Instructional methods will primarily consist of lectures and in-class discussions, however other methods such as collaborative projects and student teaching may be adopted according to the particular needs of the class.

Course Objectives

Students should be able to demonstrate competency in the techniques and topics outlined in the section above. Additionally students should develop general problem solving skills and apply tools acquired in the course to interpret and solve problems.

Course Materials

Textbook: *Linear Algebra: A Geometric Approach, Second Edition*

Calculator (recommended): TI-83, TI-83+, or TI-84+. Any other calculator model must be approved by the instructor.

Students should bring their calculator and textbook to class.

Grades

Grades will be taken from homework, quizzes, and Exams as follows:

Grading scale: $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

Evaluation: Quizzes = 20%; Exams 1, 2, and 3 = 20% each; Final Exam = 20%.

Quizzes: Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, take home assignments, and in-class projects. The lowest quiz grade will be dropped.

Final Exam: The final Exam will be a comprehensive in-class exam; see the exam schedule jcsu.edu/happenings/exam-schedule. If beneficial, the final exam score will replace the lowest exam score.

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence.

Missed quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late take-home quizzes will have 25% deducted for each day it is late.

Missed exams: If a student misses an exam for a verifiable excuse, they should send me an e-mail *as soon as possible* so that a make-up can be arranged. Make-up exams may be significantly more difficult. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services ,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

www.jcsu.edu/student/office-of-counseling-services,
phone (704) 378-1044.

Course Content

This is an outline of topics that will be covered in this course. Additional sections may be covered as time allows

Chapter 1: Vectors and Matrices

1. Vectors
2. Dot Product
3. Hyperplanes in $\mathbb{R}^n$
4. Systems of Linear Equations
5. Theory of Linear Systems
6. Applications (if time allows)

Chapter 2: Matrix Algebra

1. Matrix Operations
2. Linear Transformations
3. Inverse Matrices
4. Elementary Matrices
5. The Transpose

Chapter 3: Vector Spaces

1. Subspaces of $\mathbb{R}^n$

2. The Four Fundamental Subspaces
3. Linear Independence and Basis
4. Dimension and Its Consequences

Chapter 4: Projections and Linear Transformations

1. Inconsistent Systems and Projections
2. Orthogonal Bases
3. The Matrix of a Linear Transformation

Chapter 5: Determinants

1. Properties of Determinants
2. Orthogonal Bases

Chapter 6: Eigenvalues and Eigenvectors

1. The Characteristic Polynomial
2. Diagonalizability
4. The Spectral Theorem

Important Dates

Wednesday 11 th September	Last day to Add/Drop
Wednesday 18 th September	Exam 1
Wednesday 23 rd October	Exam 2
Wednesday 27 th November	Exam 3
TBA	Final Exam

With exception of the final, the above exam dates are subject to change. JCSU's Fall 2012 academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: jcsu.edu/happenings/exam-schedule.

While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.